

Configuration Document for SPOT NPEWS

Introduction

This document forms configuration information for systems using the National PEWS standard as outlined in the digital specification. This information is designed to be complementary to those specifications to allow suppliers to know what settings to use for various areas. These configuration settings may be updated by NHS England as more information and research is gathered. Any update will be communicated to suppliers as part of our regular update process.

Parameters for scoring Age

Bracket: **0 – 11months**

Respiratory Rates:

Under 10 – Pink – Scores 4

10 – 19.99 – Orange – Scores 2

20 – 29.99 – Yellow – Scores 1

30 – 49.99 – White – Scores 0

50 – 59.99 – Yellow – Scores 1

60 – 69.99 – Orange – Scores 2

70 and above – Pink – Scores 4

Respiratory Distress:

Mild – Yellow – Scores 1:

- Nasal flaring
- Subcostal recession

Moderate – Orange – Scores 2:

- Head bobbing
- Tracheal tug
- Intercostal recession
- Inspiratory or expiratory noises

Severe – Pink – Scores 4:

- Sternal recession
- Grunting
- Exhaustion
- Impending respiratory arrest

Oxygen Saturation:

Below or equal to 91% - Pink – Scores 4

91.01% to 94.99% - Yellow – Scores 1

Above or equal to 95% - White – Scores 0

Oxygen Support device:

Pink – Scores 4 and overrides any score from oxygen support level:

- HF – High Flow
- BiP – BiPAP
- CP – CPAP

White – Scores 0 and allows the score from oxygen support to be used:

- NP – Nasal prongs
- FM – Face mask
- HB – Head box
- NRB – Non Rebreather

Oxygen Support level – Percentages:

Air / No Support – White – Scores 0

16% to 29.99% – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

30% to 49.99% – Orange – Scores 2

50% or higher – Pink – Scores 4

Oxygen Support level – Litres per min:

Air / No Support – White – Scores 0

0.01 to 1.99 – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

2 to 5.99 – Orange – Scores 2

6 or higher – Pink – Scores 4

Heart Rate:

Under 80 – Pink – Scores 4

80 – 89.99 – Orange – Scores 2

90 – 109.99 – Yellow – Scores 1

110 – 149.99 – White – Scores 0
150 – 169.99 – Yellow – Scores 1
170 – 179.99 – Orange – Scores 2
180 and above – Pink – Scores 4

Blood Pressure (systolic):

Under 50 – Pink – Scores 4
50 – 59.99 – Orange – Scores 2
60 – 69.99 – Yellow – Scores 1
70 – 89.99 – White – Scores 0
90 – 99.99 – Yellow – Scores 1
100 – 109.99 – Orange – Scores 2
110 and above – Pink – Scores 4

Capillary Refill time:

Below or equal to 2 seconds – White – Scores 0
Equal to or above 3 seconds – Orange – Scores 2

Age Bracket: **1 – 4 years** Respiratory

Rates:

Under 10 – Pink – Scores 4
10 – 19.99 – Orange – Scores 2
20 – 39.99 – White – Scores 0
40 – 49.99 – Yellow – Scores 1
50 – 59.99 – Orange – Scores 2
60 and above – Pink – Scores 4

Respiratory Distress:

Mild – Yellow – Scores 1:

- Nasal flaring
- Subcostal recession

Moderate – Orange – Scores 2:

- Head bobbing
- Tracheal tug
- Intercostal recession
- Inspiratory or expiratory noises

Severe – Pink – Scores 4:

- Sternal recession
- Grunting
- Exhaustion
- Impending respiratory arrest

Oxygen Saturation:

Below or equal to 91% - Pink – Scores 4

91.01% to 94.99% - Yellow – Scores 1

Above or equal to 95% - White – Scores 0

Oxygen Support device:

Pink – Scores 4 and overrides any score from oxygen support level:

- HF – High Flow
- BiP – BiPAP
- CP – CPAP

White – Scores 0 and allows the score from oxygen support to be used:

- NP – Nasal prongs
- FM – Face mask
- HB – Head box
- NRB – Non Rebreather

Oxygen Support level – Percentages:

Air / No Support – White – Scores 0

16% to 29.99% – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

30% to 49.99% – Orange – Scores 2

50% or higher – Pink – Scores 4

Oxygen Support level – Litres per min:	2
2 to 5.99 – Orange – Scores	2
6 or higher – Pink – Scores	2
Heart Rate: Under 60 – Pink – Scores	7

Air / No Support – White – Scores 0

0.01 to 1.99 – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

60 – 69.99 – Orange – Scores 2 70

– 89.99 – Yellow – Scores 1

90 – 139.99 – White – Scores 0

140 – 149.99 – Yellow – Scores 1

150 – 169.99 – Orange – Scores 2

170 and above – Pink – Scores 4

Blood Pressure (systolic):

Under 50 – Pink – Scores 4

50 – 59.99 – Orange – Scores 2

60 – 79.99 – Yellow – Scores 1

80 – 99.99 – White – Scores 0

100 – 119.99 – Yellow – Scores 1

120 – 129.99 – Orange – Scores 2

130 and above – Pink – Scores 4

Capillary Refill time:

Below or equal to 2 seconds – White – Scores 0

Equal to or above 3 seconds – Orange – Scores 2

Age Bracket: **5 – 12 years** Respiratory

Rates:

Under 10 – Pink – Scores 4

10 – 14.99 – Orange – Scores 2

15 – 19.99 – Yellow – Scores 1

20 – 24.99 – White – Scores 0

25 – 39.99 – Yellow – Scores 1

40 – 49.99 – Orange – Scores 2

50 and above – Pink – Scores 4

Respiratory Distress:

Mild – Yellow – Scores 1:

- Accessory muscle use

Moderate – Orange – Scores 2:

- Tracheal tug
- Intercostal recession
- Inspiratory or expiratory noises

Severe – Pink – Scores 4:

- Tripoding
- Supraclavicular recession
- Grunting
- Exhaustion
- Impending respiratory arrest

Oxygen Saturation:

Below or equal to 91% - Pink – Scores 4

91.01% to 94.99% - Yellow – Scores 1

Above or equal to 95% - White – Scores 0

Oxygen Support device:

Pink – Scores 4 and overrides any score from oxygen support level:

- HF – High Flow
- BiP – BiPAP
- CP – CPAP

White – Scores 0 and allows the score from oxygen support to be used:

- NP – Nasal prongs
- FM – Face mask
- HB – Head box
- NRB – Non Rebreather

Oxygen Support level – Percentages:

Air / No Support – White – Scores 0

16% to 29.99% – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

30% to 49.99% – Orange – Scores 2

50% or higher – Pink – Scores 4

Oxygen Support level – Litres per min:

Air / No Support – White – Scores 0

0.01 to 1.99 – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

2 to 5.99 – Orange – Scores 2

6 or higher – Pink – Scores 4

Heart Rate:

Under 60 – Pink – Scores 4

60 – 69.99 – Orange – Scores 2

70 – 79.99 – Yellow – Scores 1

80 – 119.99 – White – Scores 0

120 – 139.99 – Yellow – Scores 1

140 – 159.99 – Orange – Scores 2

160 and above – Pink – Scores 4

Blood Pressure (systolic):

Under 70 – Pink – Scores 4

70 – 79.99 – Orange – Scores 2

80 – 89.99 – Yellow – Scores 1

90 – 109.99 – White – Scores 0

110 – 119.99 – Yellow – Scores 1

120 – 129.99 – Orange – Scores 2

130 and above – Pink – Scores 4

Capillary Refill time:

Below or equal to 2 seconds – White – Scores 0

Equal to or above 3 seconds – Orange – Scores 2

Age Bracket: **13 – 18 years**

Respiratory Rates:

Under 10 – Pink – Scores 4

10 – 14.99 – Yellow – Scores 1

15 – 24.99 – White – Scores 0

25 – 29.99 – Yellow – Scores 1

30 – 39.99 – Orange – Scores 2

40 and above – Pink – Scores 4

Respiratory Distress:

Mild – Yellow – Scores 1:

- Accessory muscle use

Moderate – Orange – Scores 2:

- Tracheal tug
- Intercostal recession
- Inspiratory or expiratory noises

Severe – Pink – Scores 4:

- Tripoding
- Supraclavicular recession
- Grunting
- Exhaustion
- Impending respiratory arrest

Oxygen Saturation:

Below or equal to 91% - Pink – Scores 4

91.01% to 94.99% - Yellow – Scores 1

Above or equal to 95% - White – Scores 0

Oxygen Support device:

Pink – Scores 4 and overrides any score from oxygen support level:

- HF – High Flow
- BiP – BiPAP
- CP – CPAP

White – Scores 0 and allows the score from oxygen support to be used:

- NP – Nasal prongs
- FM – Face mask
- HB – Head box
- NRB – Non Rebreather

Oxygen Support level – Percentages:

Air / No Support – White – Scores 0

16% to 29.99% – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

30% to 49.99% – Orange – Scores 2

50% or higher – Pink – Scores 4

Oxygen Support level – Litres per min:

Air / No Support – White – Scores 0

0.01 to 1.99 – Yellow – Scores 1 (Note if your system allows support under this level, it should still score 1)

2 to 5.99 – Orange – Scores 2

6 or higher – Pink – Scores 4

Heart Rate:

Under 50 – Pink – Scores 4

50 – 59.99 – Orange – Scores 2

60 – 69.99 – Yellow – Scores 1

70 – 99.99 – White – Scores 0

100 – 119.99 – Yellow – Scores 1

120 – 129.99 – Orange – Scores 2

130 and above – Pink – Scores 4

Blood Pressure (systolic):

Under 80 – Pink – Scores 4

80 – 89.99 – Orange – Scores 2

90 – 99.99 – Yellow – Scores 1

100 – 119.99 – White – Scores 0

120 – 129.99 – Yellow – Scores 1

130 – 139.99 – Orange – Scores 2

140 and above – Pink – Scores 4

Capillary Refill time:

Below or equal to 2 seconds – White – Scores 0

Equal to or above 3 seconds – Orange – Scores 2

Options for skipping an observation

Blood Pressure:

1. Not Attempted (No Concern) (NCO) – Scores 0.
2. Unsuccessful Attempt (No Concern) (U0) – Scores 0.
3. Unsuccessful attempt (Concern) (U4) – Scores 4.
4. Other (Text Box to complete) – user selects whether to score 0 or 4.

Carer Question

1. Skipped (no reasons presented)

Clinical Intuition

1. Skipped (no reasons presented)

Short codes for Respiratory Support Devices

HF – High Flow

BiP – BiPAP

CP – CPAP

NP – Nasal prongs

FM – Face mask

HB – Head box

NRB – Non-rebreather

Short codes for escalation reason

SC – Specific Concern

CQ – Carer Question

CI – Clinical Intuition

P – PEWS

0 – None

Escalation options for Clinical intuition

Yes – Concerned

No – Not Concerned

If “Yes” is selected, they then have the option to escalate to any level:

None

Low

Medium

High

Emergency

Escalation options for Parent / Carer question

W – Worse

S – Same

B – Better

A – Parent/Carer Asleep

U – Unavailable

If “Worse” is selected, they then have the option to escalate to any level:

None

Low

Medium

High

Emergency

Escalation Guidance

Escalation Level: **Low**

Specific Concern: None

Clinical Intuition: Nurse/Clinician concern that patient needs increased monitoring despite low PEWS

Carer Question: Carer uses words that suggests the child needs increased monitoring or intervention despite the low PEWS

PEWS Score: 1-4

Communication and Response: Inform Nurse in Charge, Consider medical review by ST3+ or equivalent, Bedside Nurse to feedback plan to parents

Medical Review Timings: As agreed with medical team

Minimal Observations: Must reassess within 60 minutes (and then document ongoing plan)

Escalation Level: **Medium**

Specific Concern: New suspicion of sepsis

Clinical Intuition: Nurse/Clinician concern that patient needs a medical review irrespective of PEWS

Carer Question: Carer uses words that suggests the child needs a clinical review irrespective of

PEWS PEWS Score: 5-8

Communication and Response: Review by Nurse in charge for potential escalation (and/or Outreach Nurse or equivalent), Request Medical Review by ST3+ or equivalent, Stabilisation plan to be considered, Bedside Nurse to feedback to parents

Medical Review Timings: Within 30 minutes

Minimal Observations: Must reassess within 30 minutes (and then document ongoing plan), Continuous Oxygen Saturation Monitoring needed

Escalation Level: **High**

Specific Concern: AVPU: Change to AVPU – V “Responsive only to Voice” or New suspicion of septic shock

Clinical Intuition: Nurse/Clinician concern that patient needs a 'Rapid Review' irrespective of PEWS

Carer Question: Carer uses words that suggests the child needs a 'Rapid Review' irrespective of

PEWS PEWS Score: 9-12

Communication and Response: Immediate review by Nurse in Charge for potential escalation, Call for

'Rapid Review': Medical incl. airway skills ST3+ and outreach nurse (If available or equivalent),
Stabilisation plan to be discussed with consultant, Senior nurse to feedback plan to parents,

Medical Review Timings: Within 15 minutes

Minimal Observations: Every 30 minutes and continuous monitoring of Respiratory Rate / Oxygen Saturation / ECG, GCS recording if change in AVPU

Escalation Level: **Emergency**

Specific Concern: Change to AVPU – P or U "Responsive only to Pain" or "Unresponsive" or
Abnormal pupillary response

Clinical Intuition: Nurse/Clinician concern that patient needs emergency review for life-threatening situation

Carer Question: Carer uses words that suggests the child has collapsed or significantly deteriorated

PEWS Score: ≥ 13

Communication and Response: Immediate 2222 call: "Paediatric Medical Emergency" and review by Nurse in Charge, Consultant informed urgently to confirm stabilisation plan, Senior nurse to support and feedback to parents, [In specialist environments rapid review can replace 2222 but only with prior agreement between consultant and Nurse in Charge].

Medical Review Timings: Immediate

Minimal Observations: Every 15 minutes and continuous monitoring of Respiratory Rate / Oxygen Saturation / ECG, GCS recording if change in AVPU or abnormal pupillary response

Sepsis Guidance

THINK! Could this be sepsis?

Think sepsis if any of the following are present:

- Neutropenia or immunocompromised (call medical professional for immediate review)
- Known or suspected infection
- Temperature $\geq 38^{\circ}\text{C}$ or $< 36^{\circ}\text{C}$
- Increasing oxygen requirement

- Unexplained tachypnoea/ tachycardia
- Altered mental state (e.g. lethargy/floppy)
- Prolonged CRT, mottled or ashen appearance

If suspicion of sepsis, inform nurse in charge. Escalate to patient's own or on-call team. **ISBAR**

Guidance

I – Hello, I am staff nurse (xx) from Ward (xx), I am calling about (xx).

S – I am calling because (e.g. PEWS increased to xx, carer is concerned because xx). The last observations were (xx).

B – They are (age), admitted on (date) for (reason). They recently had surgery (xx); treatment (xx).

A – I think they are (e.g. hypovolaemic). I don't know what is wrong with them but I am/carers is very concerned.

R – I would like you to (e.g. review in xx minutes please).